

Prosody and Information Structure

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Keywords

focus, accent, prominence, givenness, intonation

Abstract

In speech, information is conveyed using words, syntactic structures, and prosody to distinguish new information from discourse-given or inferable information and link the meanings of words and phrases to discourse antecedents. This review examines the role of prosody in communicating information structure. The point of departure is seminal work on information structure developed with reference to English, and corresponding work laying the foundation for current approaches to prosody as a phonological phenomenon. Corpus and experimental studies are reviewed for evidence that (a) speakers produce prosody in relation to focus and givenness and (b) listeners perceive and interpret prosodic cues to information structure meaning. Empirical findings show qualified evidence that listeners attend to prosody in processing and comprehending speech, though production data clearly show a many-to-many correspondence between form and meaning. A proposal that bridges these findings relates phonological and/or phonetic prominence scales to scales over information structure.

1. THE COMPONENTS OF INFORMATION STRUCTURE

Theme: content of a sentence that conveys what the sentence is about

Rheme: content of a sentence that predicates information about the theme

Focus: a focused word or phrase is interpreted relative to alternative expressions that are salient in the prior discourse

Speakers structure their sentences and render them in spoken form in ways that relate the informational content of a sentence to its discourse context, for example, distinguishing new information from information that is already given. This phenomenon, understood to be a universal property of human language, is termed information structure. Many languages use prosody to mark information structure through variation in patterns of word-level prominence and in prosodic phrasing. This review examines the relationship between prosody and information structure as viewed from the intersecting perspectives of phonology, phonetics, and pragmatics.

The information content of the words and phrases in a sentence is defined in relation to the speaker's and hearer's knowledge or beliefs as updated by the prior discourse context (Féry & Ishihara 2016). Information structure has been described in terms of several properties, including theme/rheme, focus/background, topic/comment, and old/new, properties that characterize patterns of prosodic prominence along with pronominalization, syntactic focus constructions, topicalization, and word order, among other phenomena. The present review concentrates on the role of prosody in conveying information structure.

Although information structure is common to all languages (Roberts 2012), there are differences among languages in the use of lexical, syntactic, and/or prosodic means to convey information structure and in the types of information structure distinctions that are conveyed through those means (É Kiss 1995, Gussenhoven 2004, Kügler & Calhoun 2021). Among languages where prosody plays a role, there are differences in how prosody interacts with lexical and syntactic choices in marking information structure as well as differences in what aspect of prosody is important for this function. Kügler & Calhoun (2021) identify three types of systems that differ in the nature of the prosodic cues to information structure: stress-based, phrase-based, and register-based.¹ Well-known examples of languages using stress-based cues include English and other West Germanic languages. Languages that use prosodic phrasing to encode information structure distinctions include French, Japanese, and Korean. Kügler and Calhoun, following Xu (1999) and others, describe Mandarin as using register-based cues, with expansion of the fundamental-frequency (F0) range on focused words and postfocal compression. In yet other languages, prosody appears to play no direct role in conveying information structure, for instance, in certain varieties of Malay (Kaland et al. 2023, Maskikit-Essed & Gussenhoven 2016) and in diverse African languages (Downing & Hyman 2015, Güldemann et al. 2015).

Much of the seminal work on information structure was conducted with reference to English, a language in which prosody alone or in conjunction with lexical and syntactic means plays a key role in encoding information structure. English, along with other West Germanic languages, is also frequently featured in contemporary experimental work. Issues that arise in the analysis of English, concerning the number and types of categories that structure the relationship between prosody and information structure, the representational and grammatical devices that mediate their relationship, and methods for evaluating empirical evidence for prosodic encoding of information structure, also arise in the analysis of other languages. Recognizing the practical limits on the content of this review, I have chosen to focus on phenomena first described and analyzed with reference to English. This means that many interesting and important phenomena in other languages are excluded or receive only brief mention. In addition to the comparative and typological studies cited above, other reviews of prosody and information structure are suggested that the interested reader may consult for added breadth and depth on topics that are insufficiently treated here (see the Related Resources following the Literature Cited).

¹For references to primary sources on languages exemplifying each type of system in their tripartite typology, readers are referred to Kügler & Calhoun (2021).

1.1. Information Packaging and the Given/New Distinction

Scholarly work on information structure has its roots in linguistic functionalism, in particular in the work of Vilém Mathesius, founder of the Prague School. Mathesius [1975 (1961)] describes sentences in terms of two components: theme (what is being spoken about) and nucleus (what is being said about it; appears as “rheme” in later English translations). These foundational concepts were further developed in Firbas’s (1992) Functional Sentence Perspective, centered on the notion of communicative dynamism: the relative contribution that a linguistic form makes to the ongoing communication. In this approach, the speaker’s communicative intentions drive linguistic choices related to word order, sentence structure, intonation, and the contextual “recoverability” of the linguistic elements of a sentence; collectively, these properties determine the communicative dynamism of the sentence or utterance. Prague School functionalism is also reflected in the work of Daneš (1960; see also Daneš 1987) with emphasis on the role of intonation in marking the parts of a sentence that contribute new information to the discourse.

The term “information structure” is attributed to Halliday (1967) in work that builds further on the functionalist approach of the Prague School. Halliday describes speech as being organized in “information units,” each containing one element that is intonationally marked as the information focus. For Halliday, information focus marks the part of the message that is new to the discourse, while linguistic material outside of the information focus is given. For example, in the sentence *The teacher is **late***, the information focus is on the prosodically marked word (in bold), with *the teacher* taken as given information. Halliday relates the given/new distinction to the concepts of theme and rheme from Functional Sentence Perspective (Firbas 1991). In English, the information focus can be located anywhere in the information unit, but the default location is sentence-final position, by which given information precedes new. Information focus in nonfinal position, for instance, ***The teacher** is late*, marks new information that is contrastive.

“Information status” is the term widely used in reference to Halliday’s given/new distinction. Chafe (1976) interprets the given/new distinction in relation to consciousness (Chafe 1974; see also Lambrecht 1994). Given information is cognitively activated, expressing knowledge that the speaker assumes to be in the consciousness of the addressee at the time of the utterance, while new information is that which the speaker’s utterance makes newly activated in the mind of the addressee. Chafe’s thesis is that a speaker plans their utterance to accommodate the mental state of their addressee, determining the propositional content and packaging information accordingly using lexical and syntactic devices and prosody. Under Chafe’s proposal, information packaging calls for the speaker’s reckoning with the knowledge state of the addressee. This relates to the notion of Common Ground (Lewis 1979; Stalnaker 1974, 2002) as information that is mutually recognized by interlocutors as shared knowledge or belief, specific to the communication situation.

Subsequent proposals further elaborate the notion of givenness. Clark & Haviland (1977) emphasize its centrality to Grice’s (1975) cooperative principle, while Prince (1981) emphasizes the distinction between givenness based on the explicit content of prior discourse and givenness based on a speaker’s assumptions about information that the hearer can predict or infer from sentence and discourse context, or information that the hearer already knows. Baumann & Riester (2012) further expand the notion of givenness to distinguish lexical givenness (i.e., prior mention) from referential givenness.

Givenness: a given element (word or phrase) has an overt or inferable antecedent in the prior discourse context; a new element introduces a new entity

1.2. Focus as Alternatives

Focus is another dimension of information structure and has been intensively studied in relation to prosody. Rooth (2016) defines focus as a grammatically mediated relationship between a phonology–phonetics of prominence and semantic–pragmatic factors that relate the

Narrow focus:

focused material that answers the question under discussion

Broad focus: focus that extends over an entire sentence

Deaccenting: the suppression of phrasal stress and pitch accent on words following narrow or contrastive focus in the same prosodic phrase

Contrastive focus: focus with alternatives that are overt in the discourse context, graded by degree of similarity of alternatives and focused word

interpretation of a linguistic expression (word or phrase) to alternative expressions. Two meaning functions have been associated with focus (Krifka 2008). A semantic function occurs in sentences with focus-sensitive operators like English *only*, *also*, or *even*, whose semantic value in a sentence depends on the linguistic expression associated with focus or its denotation (Roberts 2012, Rooth 1992). Pragmatic functions of focus relate to interlocutors' management of the Common Ground (Krifka 2008), for instance, establishing that an answer to a question appropriately updates the Common Ground—this is termed Question–Answer (Q–A) congruence. Other pragmatic functions of focus include confirmation and correction. For example, following the statement *Sue went with Bill*, an interlocutor may respond with (*Yes*,) *Sue_F went with Bill*, where focus on *Sue* confirms information about who went with Bill, while the response (*No*,) *Mary_F went with Bill*, with focus on *Mary*, corrects the same information. An alternative approach conflates the distinction between semantic and pragmatic functions of focus under a strictly pragmatic account, where focus marks the linguistic expression that is the answer to a current discourse goal, formulated as a question under discussion (QUD) (Büring 2003, Ginzberg 2012, Roberts 2012, Velleman & Beaver 2016). Viewed in this way, every assertion can be understood as the answer to a question, whether the question is explicitly posed or implicit with respect to the prior discourse. Thus, the sentence *Mary sent flowers_F* can be used to answer the explicitly stated QUD *What did Mary send?* but is also appropriate in the absence of an explicit question when the same QUD can be inferred from the situation. The term narrow focus is often used to refer to this pragmatic function, where the focused expression provides the answer to an explicit or implicit QUD.

The effect of focus is understood as enriching the meaning of the focus-marked word or phrase by invoking a set of contrastive semantic alternatives that are salient (i.e., cognitively activated) due to being explicitly mentioned, inferred, or otherwise accessible given the discourse context (Bolinger 1958, 1961, 1972; Chafe 1974, 1976; Halliday 1967). Thus, in the sentence *Mary_F went with Bill*, the word *Mary* is to be interpreted in relation to alternative expressions such as *Sue went with Bill*, *My brother went with Bill*, etc. This notion of focus is formalized in Rooth's (1992) theory of Alternative Semantics, where focus is defined in terms of three parameters: the location of the focused word in the sentence, the scope of focus on a constituent containing the focused word, and the antecedent of focus in the shared knowledge of the interlocutors. The location parameter reflects the fact that focus is not positionally restricted and can be associated with any word in the sentence. The scope parameter relates to the long-standing observation that focus can be associated with a single word or a larger syntactic constituent to which it belongs (Halliday 1967). For example, focus that is prosodically realized on the object noun, as in *Mary sent flowers_F*, can be interpreted as having narrow focus associated with the noun [*flowers*]_F or with the verb phrase [*sent flowers*]_F or broad focus associated with the entire sentence [*Mary sent flowers*]_F, where brackets delimit the syntactic scope of focus. Focus must have scope over the constituent that satisfies Q–A congruence, which has consequences for the prosodic expression of focus. In Rooth's formulation, the semantic value of a focused word is the open expression formed by replacing the focused word or phrase with alternatives of the same semantic type. This replacement results in an alternative set for the focused word. For instance, the semantic value of *Bill* in *Mary went with [Bill]_F* is the set of propositions obtained by replacing *Bill* with semantic alternatives *x* in *Mary went with x*. The antecedent parameter under Rooth's analysis, as in earlier formulations of focus, requires that the focused expression have at least one antecedent in the discourse that is in its alternative set. Subsequent proposals introduce additional restrictions on the antecedent condition to account for patterns of deaccenting in English (Büring 2019, Wagner 2005).

Contrastive focus occurs when the focused word has an overt alternative in the discourse context and may be graded based on the similarity of overt alternatives to the focused word (Repp 2016). One such context involves coordination constructions; for instance, a discourse context

with the statement *It was Larry or Jill who sent flowers* followed by the response *[Jill]_F sent flowers* conveys a contrastive focus interpretation for *[Jill]_F* where *Larry sent flowers* is a member of the alternative set. Another such context involves correction of an explicit antecedent. In addition, some authors recognize a distinction between contrast associated with the topic or theme of a sentence (contrastive topic) and contrast within the comment or rheme portion, claiming different intonational patterns that manifest in different pitch contours (Bolinger 1965, Büring 2016, Jackendoff 1972, Steedman 2014).

To summarize, we have discussed two main information structure distinctions that may be prosodically expressed, givenness and focus, with additional distinctions—for instance, in the cognitive activation of lexical expressions and their denotations, in the status of a discourse antecedent of a word or phrase as explicit versus implicit or inferable, and in givenness as categorical versus gradient. Related to focus and semantic alternatives are distinctions in the location of focus in sentence theme/topic versus rheme/comment and in the scope of focus. We return to examining the evidence for prosodic correlates of these information structure distinctions in Section 3, following a brief review of prosody in its phonological and phonetic form.

Before moving on, it bears noting that while this review looks at work on information structure related to givenness, focus, and (to a lesser extent) topic, these do not exhaust the kinds of relationships that may hold between the informational content of a sentence and its discourse context. Even within these categories, there are rich and varied distinctions, for instance, among types of focus as described for English (Repp 2016) and in other languages using syntactic, lexical, and prosodic means of encoding (É Kiss 1995, Li & Thompson 1976). Furthermore, outside the theoretical frameworks discussed here, other types of discourse relations related to information structure have been proposed. For example, Rhetorical Structure Theory (RST) (Mann & Taboada 2025, Mann & Thompson 1988) recognizes an open-ended set of discourse relations that includes Background, Elaboration, Cause, Result, Motivation, Purpose, and Contrast, among others. RST has been used for the creation of language databases with discourse annotation (Carlson et al. 2002) and in work developing speech technologies, for instance, using prosodic cues to recognize discourse relations (Kleinhanz et al. 2017) or synthesizing phrasal prosody based on discourse relations detected from text (Hu et al. 2016).

2. PROSODIC FEATURES RELATED TO INFORMATION STRUCTURE

Prosody refers to the melody, rhythm, stress, and phrasing of speech, which in linguistic approaches relate to phonological structure. Syllables are grouped into metrical feet, and words are grouped into prosodic phrases. Both structures recognize one position (initial or final) as strong, that is, prominent relative to the other, weak elements in the structure (Ladd & Arvaniti 2023). In so-called stress languages like English, strong positions determine the location of stress at the word and phrase levels.

The term “intonation” is used interchangeably with “prosody” in analyses focused on pitch patterns at the phrase level. Emerging from earlier impressionistic descriptions (e.g., Bolinger 1958, O'Connor & Arnold 1973, Pike 1945), a phonological approach informed by acoustic observations of American English pitch trajectories was proposed by Pierrehumbert (1980) and ultimately codified in Autosegmental-Metrical (AM) theory (Ladd 2008). AM theory builds on the idea from Liberman (1975) that the intonational tones associated with words in sentence contexts are grouped such that within each grouping, one tone element (designated as the nucleus, or head element) has greater prominence than other tonal elements in the same grouping.² In the

Contrastive topic: a focused theme or topic relates the meaning of a sentence to alternative topics or alternative questions (e.g., *Who did what?*)

Autosegmental-Metrical (AM) theory: a phonological theory of intonation that groups words into prosodic phrases with tone autosegments marking phrase boundaries and phrasal stress

²Ladd (1986) discusses related proposals predating AM theory, by Chao (1932), Pike (1945), and O'Connor & Arnold (1973).

Pitch accent: a tonal feature associated with a word that has phrasal stress (also sometimes called simply “accent”)

Default phrasal stress: phrasal stress assigned on structural grounds, in the absence of factors such as focus that force stress alignment to other positions

AM model of English, these groupings are identified as prosodic phrases, defined at two levels in a hierarchical layering of prosodic constituent structure (Nespor & Vogel 2007). The lower level is the intermediate phrase, which is the domain in which obligatory phrasal stress is assigned.³ The default location for phrasal stress is the rightmost stressable word in the intermediate phrase. In longer phrases, metrical structure may license phrasal stress in additional locations, but only the rightmost phrasal stress, termed the nuclear stress, is obligatory. The analysis of prosodic phrasing relates to the analysis of information structure in English because the location of nuclear stress and its association with tonal features (pitch accents) depend on information structure: Whereas the default phrasal stress pattern locates nuclear stress on the phrase-final word, in sentences with a narrow or contrastive focus on an earlier word, nuclear stress will be located earlier, on the focused word (see discussion in Section 3.1). In such cases, words following nuclear stress are ineligible for phrasal stress.

The higher level of prosodic phrase is the intonational phrase, which is composed of one or more intermediate phrases, each with an obligatory phrase-level nuclear stress. Depending on the length and syntactic complexity of a sentence, it may be produced as one or more intonational phrases, within which there may be one or more intermediate phrases.

It bears noting that the AM treatment of prosodic phrasing has been challenged on multiple fronts related to its adherence to a strictly two-level distinction between intermediate and intonational phrases. In the AM analysis, based on work by Pierrehumbert (1980) and Beckman & Pierrehumbert (1986), phrase structure is hierarchically layered and nonrecursive. Yet observations of more finely graded distinctions in the strength of prosodic phrase juncture present a challenge for this two-level model, with some works calling for recursive layering of a unitary prosodic phrase type (Ladd 1986, 1988, 2008; Wagner 2010) and others suggesting a continuous-valued boundary strength (Krivokapić & Byrd 2012, Mooshammer et al. 2025). While it is important for future work to reconcile prosodic theory with analyses of recursive phrasing and/or continuous-valued prosodic phrase boundaries, any such proposal must preserve the central role of the prosodic phrase as the domain for phrasal stress assignment and pitch accenting. We leave that as a priority for future work and turn now to the phenomenon of accentuation.

AM theory posits an inventory of tone features: pitch accents associated with phrasal stress, and edge tones that align with the right edge of prosodic phrases. Edge tones in turn comprise phrase accents marking the end of an intermediate phrase, and boundary tones marking the end of an intonational phrase. A pitch accent is obligatory on the nuclear stressed word and is optional in other locations of phrasal stress. Prosodic phrase structure and associated tone features are illustrated in example 1. Bolded syllables have phrasal stress, with nuclear stress in bold capital letters. Tone features are H(igh) and L(ow). Starred tones are pitch accents, L- indicates phrase accents, L% indicates a boundary tone, ip indicates an intermediate phrase, and IP indicates the intonational phrase:

- (1) H* L- H* !H* L-L%
[[the **journalist's** **ES**say]_{ip} [reflected her **personal** ex**PER**ience]_{ip}]_{IP}

In AM theory, pitch accents, phrase accents, and boundary tones are viewed as discrete phonological categories, which, like segmental features, can convey categorical distinctions in meaning. Adopting this view, Pierrehumbert & Hirschberg (1990) propose a compositional account of

³ In languages that do not operate with word-level stress (e.g., Japanese, French, Hindi–Urdu, Korean), the AM framework has the accentual phrase (AP) as the smallest prosodic domain, typically spanning one or two words. Within the AP, tones are lexically specified (in pitch accent languages like Japanese) or inserted by default (e.g., in French, Hindi–Urdu, and Korean) (Jun 2014, chapter 17).

intonational meaning for English in which pitch accents convey distinctions in information structure. Their proposal assumes an inventory of five pitch accent features: L*, H*, L+H*, L*+H, and H+!H*, plus !H* (a downstepped high), where the starred tone aligns with the accented syllable (Ladd 2008, pp. 87–107; Pierrehumbert 1980). The L* accent describes a fall to a low pitch in the accented syllable, H* is a shallow rise to a moderate pitch peak on the accented syllable, L+H* has a steep rise from low pitch to an early peak on the accented syllable, L*+H rises late in the accented syllable with a high peak on the following syllable, and H+!H* is a falling accent with a downstepped high peak on the accented syllable. A categorical distinction between H* and L+H* is called into question; it is a difficult distinction to make in prosodic annotation (Pitrelli et al. 1994) and has not been empirically confirmed (e.g., Calhoun 2012, Ladd & Schepman 2003, Steffman et al. 2024). The distinction between L* and unaccented words is similarly challenging for annotators (Ladd 2022) but has not to my knowledge been tested in annotation studies or controlled experiments.

Pitch accents are phonetically implemented through dynamic pitch movements in the vicinity of an accented syllable, which correspond to time-varying F0 patterns in the acoustic signal. Because pitch accents occur on words with phrasal stress, accented syllables also manifest phonetic properties related to stress, which include longer duration and increased loudness (i.e., acoustic intensity) with hyperarticulation of segmental content.

The next section reviews evidence from the production, comprehension, and processing of pitch accents for the proposal that pitch accent categories structure the correspondence between prosody and information structure.

3. INFORMATION STRUCTURE, ACCENTING, AND PROMINENCE

The relationship between information structure and prosody has been explored in numerous studies spanning decades. This includes “impressionistic” work that is informed by researcher intuitions and auditory impressions of phrasal prosody in relation to anecdotally observed or imagined discourse contexts, and “empirical” work that is informed by audio recordings, digital images of pitch tracks overlaid on spectrograms, and acoustic measurements from those audio materials.

3.1. Information Structure and Accenting

Distinctions in information structure related to focus and givenness have been discussed in relation to the location of the nuclear pitch accent in the prosodic phrase, the tonal specification of the nuclear pitch accent, and the presence or absence of prenuclear pitch accents.

3.1.1. Nuclear accent. Impressionistic approaches are largely concerned with analyzing patterns of accenting and deaccenting in relation to focus and givenness, where “accent” is used to describe phrasal prominence, often with reference to patterns of dynamic pitch, and “deaccenting” refers to the notable absence or suppression of accent on words that may, in other contexts, be accented; deaccenting serves to highlight an accent occurring elsewhere in the prosodic phrase. Accenting and deaccenting are governed by a complex interplay of syntactic and pragmatic factors. Pragmatic factors are central in Bolinger’s work (1961, 1972, 1985), which was a major influence on the later development of AM theory (Ladd 2008, Pierrehumbert 1980). In the AM approach, accent is described in terms of prominence-lending tone features, which can be associated with any word in a sentence based on its focus or givenness status (for the relationship between phonological and information structural prominence, see Ladd & Arvaniti 2023). Other approaches relate focus to accenting via the mediating structure of phrasal stress (Chomsky & Halle 1968). Sentences where accent is associated with the default phrasal stress (final in the intermediate phrase) convey broad focus, for instance, *Mary went with **Bill*** (bolded word is accented). Accent marking a

nondefault location of phrasal stress conveys narrow or contrastive focus, for instance, *Mary went with Bill*.

Most contemporary work adopts a hybrid approach distinguishing between focus as a semantic–pragmatic construct and pitch accent as a phonological construct related to phrasal stress (Ladd 2008, pp. 217–18). A word or phrase with narrow or contrastive focus is required to align with phrasal stress, no matter its position in the sentence, and phonological rules of stress assignment fix the location of accent within the focus domain (Gussenhoven 1983a, 2007; Selkirk 1995). Thus, prompted by the question *What did Mary drive?*, a speaker may respond, *Mary drove [the red convertible]_F*, with narrow focus on the entire object NP and accent on the word with nuclear stress in the focus domain. Givenness status also comes into play in contexts where accent is shifted away from the location of default phrasal stress. For example, if the discourse context provides *Mary drove the red convertible. What did Bill drive?*, a speaker may respond with *Bill drove the blue convertible*, in which case the entire object NP *the blue convertible* has narrow focus as the answer to the question, yet accent is associated with *blue* and not, as predicted by default phrasal stress, *convertible*. In Buring’s (2019) analysis, *convertible* in the phrase *blue convertible* undergoes prominence demotion due to its status as discourse-given. Prominence demotion is implemented via stress reversal of the adjective–noun sequence (weak–strong reversed to strong–weak), which in turn forces accent to shift, constrained to associate with the position of maximum stress prominence—*blue* in this example (see also Klassen & Wagner 2017, Wagner 2005). Other examples of accent in nondefault positions of phrasal stress include topic-less “thetic” sentences with intransitive verbs where accent aligns with the subject, such as *The sky is falling* (Gussenhoven 1983a, Rochemont 2013, Sasse 1987, Schmerling 1974), and sentences where semantically light but not given words resist accent, such as *I’ve got to see a guy/someone* (Calhoun 2010). What hybrid approaches have in common is a view of accenting as a complex phenomenon bridging sound and meaning (e.g., Buring 2019; Calhoun 2009, 2010; Féry & Samek-Lodovici 2006; Gussenhoven 2004, 2007; Katz & Selkirk 2011; Schwarzschild 1999; Selkirk 1984, 1995; Steedman 1991, 2000; Truckenbrodt 2006).

3.1.2. Prenuclear accent. Most approaches to relating accent placement and focus address nuclear accent, but as noted in Section 2, words preceding the nuclear accent may also be accented. In English, prenuclear accents do not necessarily, or even typically, function to convey information structure meaning. Prenuclear accents align with positions of phrasal prominence, where most often they appear to serve a rhythmic function (Calhoun 2010, Chodroff & Cole 2018, Im et al. 2025; for German, see Baumann et al. 2021). However, in certain discourse contexts, a prenuclear accent on theme material may convey a contrastive topic or theme, for instance, *Mary_{CT} wants to kick me_F out* (Buring 2003, 2016).

3.1.3. Accent type and focus type. Information structure is conveyed not only through the location of accent but also by the type of accent; different types are associated with different information structure categories. For instance, Bolinger (1958, 1965) and Jackendoff (1972) claim that different pitch accents are used in topic/theme versus rheme position to convey focus-related meaning (see also Steedman 1991, Vallduví & Vilkuña 1998), though this is disputed by Calhoun (2012) and Martens (2022). Pierrehumbert & Hirschberg (1990) offer a detailed account of meaning associated with pitch accents in the AM model for American English, describing the use of the high-toned accent (H*) for new information, the low-toned accent (L*) for given information that does not introduce a new predication to the Common Ground, the rising accent with an early peak (L+H*) for contrastive focus, and the rising accent with a late peak (L*+H) to convey uncertainty related to a scale.

3.2. Production Evidence for Accent Distinctions Related to Information Structure

In the following sections we consider empirical evidence for proposals relating focus and givenness distinctions to phonological and acoustic correlates of accent and phrasal prominence.

3.2.1. Accent as a discrete contrast. The accented/deaccented distinction is typically treated as a discrete contrast, reflected in the literature in the use of text stylization (bold, italic, underline) for accented words. The implication is that distinctions in accent location are phonetically robust and readily recognized by speakers of the language. If this is true, prosodic annotation tasks should show high interannotator agreement on the status of words as accented or unaccented. Empirical evidence supporting this view is mixed. On one hand, several studies of prosodic annotation using the ToBI (Tones and Break Indices) system (Brugos et al. 2006) report “substantial” agreement among pairs of trained annotators on the presence or absence of pitch accent (pairwise kappa scores between 0.71 and 0.75; Bishop et al. 2020, Breen et al. 2012, Yoon et al. 2004), though Kim & Arnhold (2024) report lower agreement ($\kappa = 0.49$) for the same measure in Canadian English. In these studies, pitch accents are annotated based jointly on auditory impression and visual inspection of the pitch track from a digital display. In studies of prosody perception by ordinary listeners, interannotator agreement is much lower. For instance, several studies have used Rapid Prosody Transcription (RPT) (Cole & Shattuck-Hufnagel 2016, Cole et al. 2010), where untrained listeners assign word-level binary prominence ratings in real time based only on auditory impression. Three RPT studies report only “fair” agreement rates among large cohorts of 70–160 annotators (Fleiss multirater agreement scores between 0.21 and 0.4) (Bishop et al. 2020; Cole et al. 2010, 2019; Im et al. 2023). Their findings also show that prominence ratings are predicted not only based on the phonological and acoustic properties of a word but also based on its lexical and information structure properties and syntactic context. In other words, judged auditorily, prominence is a multidimensional construct in speech perception. It is debatable whether pitch accent annotations (ToBI: trained individuals, audiovisual judgments) or prominence annotations (RPT: untrained individuals, audio judgments) are a more appropriate comparison with the accent judgments informing impressionistic analyses of the relationship between focus and accent, but judgments made from auditory impression without the aid of visual pitch tracks are clearly closer to the way speech is processed in ordinary language use. Given the relatively low agreement reported for auditory prominence judgments, it is necessary to empirically test the specific claims about information structure and accenting that rest on such judgments. We turn next to consider such studies.

3.2.2. Accent type and unaccented words. A handful of studies on American English investigate the claim that the presence/absence and type of pitch accent depend on information structure, based on the analysis of prosody in speech production. Testing these claims in production carries the challenge of prosodic annotation, which is notoriously slow, difficult, and costly. Consequently, studies that present such data are few and small in scale. Among six such studies, including one on Canadian English (Kim & Arnhold 2024), three show a significant effect of givenness on accenting—speakers tend to produce given words as unaccented (Burdin et al. 2015, Im et al. 2023, Kim & Arnhold 2024, Turnbull 2017)—while two show no such effect (Chodroff & Cole 2019, Ito et al. 2004). Most of these studies also show an effect that focused words tend to be accented; only Chodroff & Cole’s (2019) study shows no such preference (comparable data are not available for Ito et al. 2004). Turning next to the effect of information structure on accent type, four studies show rising accents ($L+H^*$, L^*+H) to be more frequent than other accent types on words with contrastive focus (Burdin et al. 2015, Im et al. 2023, Ito et al. 2004, Turnbull 2017). There is no preferred accent type associated with contrastive focus in the other two (Chodroff &

ToBI: the Tones and Break Indices system for prosodic annotation; ToBI systems for different languages may have different tonal inventories

Cole 2019, Kim & Arnhold 2024), though we note that Chodroff & Cole's (2019) results show a significant increase in the use of high-toned pitch accents (combining H* and L+H*) in nuclear position across information structure categories: given < new < contrastive. Chodroff and Cole comment that annotator uncertainty prevented distinguishing between H* and L+H*, an annotation challenge mentioned in Section 2. Related to these studies that analyze accenting based on ToBI annotations, we note results from Ouyang & Kaiser (2015), who analyze differences in F0 shape that correspond to accent type (including unaccented) and find no significant differences between corrective focus, narrow focus, and words in broad focus contexts. These English studies can be compared with production studies on German showing a modest relationship between pitch accenting and information structure. For example, Mücke & Grice (2014) observe a probabilistic correspondence between pitch accent type and focus, similar to what is reported in two of the English studies (Chodroff & Cole 2019, Turnbull 2017), where L+H* increases in frequency across a cline of focus types—broad < narrow < contrastive—while the falling accent, H+!H*, is strongly favored in broad focus contexts (see related findings in Röhr & Baumann 2010).

Several studies have looked specifically at pitch accents in prenuclear position. Calhoun (2012) finds no difference in pitch accent type between contrastive themes versus contrastive rhemes. Chodroff & Cole (2018) also find no association between accent type and information structure (focus and givenness) for accented words in prenuclear (theme) position. Martens (2022) similarly reports no systematic differences between contrastive and noncontrastive topics and themes. The overall finding is that English speakers produce a variety of pitch accents in prenuclear position but do not systematically use different pitch accents to mark information structure-related distinctions on prenuclear themes. Related findings are reported for German with no evidence that accent type distinguishes contrastive themes and topics from noncontrastive ones (Braun 2006) or given words from those with contrastive focus (Baumann et al. 2021). Taken together, the production evidence supports the view of prenuclear accents as a primarily rhythmic phenomenon.

To summarize, the production studies from English and German present mixed evidence that nuclear pitch accents are associated with information structure. Where there are statistically significant patterns, they show a preference for given words to be unaccented and for contrastive or narrow focused words to be accented, and a preference for rising accents on words with contrastive focus. In addition, the likelihood of a rising or high accent (versus a low accent or no accent) is higher for words that are new and for words with narrow or contrastive focus. No such patterns of association are found for prenuclear accents. What accounts for the variable outcomes for nuclear accent? We might expect clearer prosodic expression of information structure in interactive tasks where interlocutors cooperate in developing the Common Ground, yet the evidence does not confirm this. For nuclear accents, among the four English studies that used interactive tasks, three show evidence for effects of information structure on pitch accent type (Burdin et al. 2015, Ito et al. 2004, Turnbull 2017), while the fourth (Ouyang & Kaiser 2015) does not. On the other hand, both German studies cited above show effects of focus on accent type, but with data from read-aloud texts in tasks like those used in the English studies by Chodroff & Cole (2019) and Kim & Arnhold (2024) that did not show such results. We might also expect stronger prosodic cues to information structure in the performative style of a TED talk speech, where speakers craft and rehearse their talks to be coherent, lively, and engaging with the audience, yet a study by Im et al. (2023) found that TED talk data showed only modest effects of information structure. In short, task differences do not go far to explain the varying outcomes across studies. The general finding is that for nuclear accent, speakers can and do produce all accent types in all information structure categories. This holds as well for prenuclear accents in English, though German speakers appear to strongly favor rising accents (L*+H, L+H*) in prenuclear position (Baumann et al. 2021). At best, there

is a probabilistic correspondence between nuclear accent type and information structure category in what is a pervasively many-to-many pattern of association.

3.3. Processing and Comprehension Evidence for Accent Distinctions Related to Information Structure

Against the findings from production data, a different picture emerges from studies comparing the effects of pitch accent type on the comprehension of information structure meaning. Several studies use a Q-A congruence paradigm where participants match questions and answers based on the focus condition of the answer as conveyed by pitch accent—for instance, matching the question *Who fried the omelette?* to the answer [**Damien**]_F *fried the omelette*, and rejecting matches with alternative answers having different focus structures, such as *Damien fried the* [**omelette**]_F. Using an interactive task, Breen et al. (2010) showed that listeners were very accurate in identifying congruent Q-A pairs based on the location of the focused word (Subject, Verb, or Object) in the answer but were much less successful in distinguishing answers that conveyed contrastive focus from those that conveyed noncontrastive focus. A study by Roettger et al. (2019), with data from a noninteractive Q-A matching task, found that listeners were only moderately successful in identifying congruent Q-A based on the prosody of the answer, with varying accuracy across the four focus conditions tested: Listeners were poor at distinguishing contrastive focus from narrow focus and distinguishing broad focus from given. For studies using the Q-A congruence paradigm to test focus domain interpretations, readers are referred to Gussenhoven (1983b) and Birch & Clifton (1995).

Other studies have investigated the online processing of information structure in a visual world paradigm, comparing the effect of accent type on reference resolution. Results show that listeners are more likely to fixate on an image depicting a new referent when hearing an accented noun, and on a given referent when hearing a deaccented noun (Dahan et al. 2002), and that L+H* accent is more likely than H* to be interpreted as a cue to contrastive focus, although both accents are compatible with a noncontrastive interpretation (Watson et al. 2008). Other studies use cross-modal priming tasks to examine the effects of pitch accent on the activation of semantic alternatives. For instance, Husband & Ferreira (2016) showed differential effects of H* and L+H* accents in priming semantic alternatives to the accented word (see also Bishop 2017; for Dutch, see Braun & Tagliapietra 2010). There is also work examining prosody in relation to scalar implicature (Chierchia 2017, Gotzner & Lacina 2024) using a variety of tasks probing the rate by which listeners calculate scalar inference. Very briefly, those studies show that the nuclear tune (pitch accent and following edge tones) plays a role in scalar inference (Gotzner & Spalek 2014, Ronai & Göbel 2024, Tomlinson et al. 2017), with the so-called rise–fall–rise (RFR) nuclear tune facilitating scalar inference to a greater extent than the falling nuclear tune. Sostarics et al. (2025) look further into the role of the pitch accent in facilitating scalar inference, comparing H*, L+H*, and L*+H accents. They find that the use of any one of these accents facilitates scalar inference in the RFR tune relative to the group of falling tunes with the same accents, with increasing rates of scalar inference in RFR across accents: H* < L+H* < L*+H. Overall, the findings from scalar implicature support a role for prosody, with an apparently categorical distinction between intonation patterns that extend over the nuclear tune (RFR versus falling) and gradient effects related to ranking over the high or rising pitch accents.

There are also studies looking specifically at the processing of prenuclear pitch accents. Studies by Gussenhoven (1983b) and Bishop (2017) report that listeners resist an interpretation of narrow or contrastive focus for a nuclear accented word in the presence of a preceding prenuclear accent—for instance, preferring an interpretation of *He **borrowed** the salt* (accents bolded, nuclear

accent underlined) as a response to *What did he do?* (VP focus) rather than *What did he borrow?* (object focus). Bishop's findings show differences among individual listeners that are partly predicted by personality traits related to autism, pointing to a role for pragmatic skills in the processing of intonational meaning. Roettger et al. (2020) report findings from English and German where participants listened to utterances in which the presence of a prenuclear pitch accent predicted the referent of the sentence-final word. Mouse-tracking data were analyzed for evidence that participants used information from prenuclear accents to anticipate and click on the referent of the sentence-final word even before it was heard in the auditory stimulus. Results show that German listeners did not use the prenuclear accent as a cue to the downstream referent, while among the English listeners, only some made predictive use of the prenuclear accent. Notably, the English prenuclear pitch accent was a larger and more salient pitch movement compared with the German one. A related German study by Braun & Biezma (2019) tested the effect of different types of prenuclear accent on listeners' interpretation of contrastive focus on the accented word, in a visual world paradigm using eye-tracking data. Gaze patterns were analyzed when the target word had a contrastive topic accent (L*+H) and compared with those where the target word had the noncontrastive prenuclear accent (L+H*). Listeners directed their gaze more to the contrastive alternative of the target word with a prenuclear contrastive (topic) accent compared with a prenuclear, noncontrastive accent. These results suggest differences between English and German in processing prenuclear accents. German listeners process prenuclear accents for information structure meaning of the accented word but appear not to integrate prenuclear accents with processing of later sentence material, while for English, some but not all listeners attend to prenuclear accent as a cue to focus-related meaning of a downstream nuclear accent. The difference between English and German may relate to the variability of prenuclear accent patterns in the two languages. German speakers strongly favor rising prenuclear accents (Baumann et al. 2021; cf. Braun 2006), while English speakers use more variable prenuclear accent patterns (Chodroff & Cole 2018).

Summarizing this section, evidence from the comprehension and processing of prosody shows an affordance for prosody to cue information structure, for at least some distinctions involving pitch accents. There are many findings that call for future investigation, listed here:

1. The effects of pitch accent on interpretation vary across tasks and depend on which comparisons among pitch accents and information structure categories are presented to listeners. It is very likely that effects are also dependent on the phonetic expression of prosody in the stimuli that are selected or created for comprehension and processing tasks, where experimenter control over phonetic variables allows salient realization of prosodic contrasts.
2. In offline tasks, listeners were accurate in identifying congruent Q-A pairs based on the location of a focus-marking accent but were less accurate in distinguishing among focus conditions based on the type of accent. Cross-modal priming as an online measure of pitch accent processing showed more consistent effects of accent type, with the L+H* pitch accent cueing contrastive focus and the presence of a prenuclear accent cueing the focus interpretation of a downstream referent.
3. For English, online processing effects revealed that individual listener differences in attention to prenuclear accents are related to pragmatic skill, suggesting individual differences in the knowledge representation of prosodic or information structure categories, or in the use of such knowledge in an experimental setting.
4. The overall weaker results for the offline Q-A congruence tasks versus online tasks may reflect perceptual accommodation. When making explicit judgments about stimuli, listeners appear willing to accommodate interpretations other than those predicted by the prosodic realization of the stimuli, for instance, routinely accepting an utterance with a narrow focus accent pattern as an answer to a question requiring a broad focus response.

5. Offline tasks probing scalar inference revealed both categorical effects at the level of the holistic nuclear tune (RFR versus falling tunes) and gradient effects among high-tone pitch accents, calling into question the status of the accent type distinction as categorical.

Acoustic prominence: acoustic properties that make a syllable or word stand out relative to other words in the local prosodic domain

3.4. Evidence for Prominence Distinctions Related to Information Structure

So far, we have considered evidence for prosodic encoding of information structure from studies that treat accent as a phonological phenomenon. Other studies examine information structure associated with prosody through the lens of acoustic prominence, bypassing the difficult challenge of annotating pitch accents and phrasal stress. Focused words in English are impressionistically described as having a higher level of prominence than nonfocused words in the same prosodic phrase (Féry & Samek-Lodovici 2006; Jackendoff 1972; Truckenbrodt 1995, 2006; Wagner 2005). Acoustic studies largely support this claim with greater acoustic prominence as measured in duration, intensity, and F0 maxima and range for words with contrastive or narrow focus compared with broad focus (Breen et al. 2010, Chodroff & Cole 2019, Cooper et al. 1985, Eady & Cooper 1986, Eady et al. 1986, Katz & Selkirk 2011, Lee et al. 2015, Xu & Xu 2005). Acoustic evidence of reduced prominence for deaccented words is less robust (Bard & Aylett 1999, Burdin et al. 2015, Calhoun 2010, Chodroff & Cole 2019, Im et al. 2023, Terken & Hirschberg 1994). Related work on German also reports variation in acoustic prominence related to information structure (e.g., Baumann & Riester 2013, Braun 2006, Féry & Kügler 2008, Roessig et al. 2022). For comparison, similar focus-related acoustic prominence and postfocal F0 compression are reported by Lee et al. (2015) for American English, Mandarin Chinese, and Standard French; the latter two languages have strikingly different prosodic systems in which pitch accents are absent (Mandarin) or serve a demarcative function in the prosodic phrase (French). Their findings suggest that focus-related prominence is not unique to languages where accenting patterns play a primary role in encoding information structure.

The general observation from the analysis of prominence in English (as in German, Mandarin Chinese, and French) is that there is a correspondence between acoustic prominence and information structure, with an increase in acoustic prominence across rankings of information structure categories related to givenness (given < accessible, inferable < new) and focus (not focused < narrow < contrastive, corrective). There is also a correspondence between acoustic prominence and pitch accent categories ranked by their tonal specification and pitch maxima, from low to high (unaccented, $L^* < H^* < L+H^* < L^*+H$; Im et al. 2023). A similar correspondence holds for German (Baumann & Röhr 2015, Baumann & Winter 2018). The perceptual prominence ratings discussed above (from RPT studies) show the same patterns of correspondence, with a higher likelihood of a (binary) prominence rating for words that have higher acoustic prominence and/or more highly ranked pitch accents, and for words that are more highly ranked on the givenness and focus scales.

These multitiered relationships between phonetic, phonological, and information structure scales suggest an explanation for the puzzling many-to-many associations between accent type and information structure categories in production discussed in Section 3.2. Contrary to widely held expectations, and to approaches that treat pitch accents as pragmatic morphemes (Bolinger 1958, Pierrehumbert & Hirschberg 1990), empirical evidence shows that all types of pitch accents may potentially appear in all information structure conditions. Does this mean that pitch accents do not, in fact, encode information structure distinctions in English? Evidence discussed above on the comprehension and processing of pitch accents suggests otherwise, but what then to make of the weak or null findings from production evidence? An account that is consistent with production evidence is that prosody relates to information structure through a mapping between prominence

scales. This is not a new claim; prominence-based accounts have been advanced by Calhoun (2010) as a model of both focus-related and rhythmic prominences, by Katz & Selkirk (2011) in an analysis of prosodic cues to contrastive focus and givenness, and by Im et al. (2023) in relating perceived prominence to pitch accent type and information structure. Yet it is a view that has not yet been widely adopted.

Under the prominence-based account, a speaker maps information structure scales ranking givenness and focus onto the pitch accent prominence scale and/or the acoustic prominence scale, or both. The choice of prosody scale may depend on other factors that govern pitch accent use, such as speaking style. Speech performed on a public stage (Im et al. 2023), broadcast speech (Yuan et al. 2005; see also Rodero 2015 for Spanish), and speech produced in an intentionally lively style (Chodroff & Cole 2018, 2019) all use more high-prominence accents (e.g., L+H*) and have an overall higher rate of accenting compared with ordinary conversational speech. There are also dialectal differences (Burdin et al. 2022) and individual speaker differences (Ouyang & Kaiser 2015) in the relative frequency of low- and high-prominence accents. With myriad factors influencing the use of pitch accents, in certain contexts a speaker may choose to express information structure distinctions through acoustic rather than accentual prominence. This prominence-based account rejects the view that pitch accents have inherent information structure meaning in English (a similar argument can be made for German) while leaving open the possibility that pitch accents have an affordance to convey information structure distinctions, for which much of the evidence relates to the use of high-prominence accents (e.g., L+H*), common in comprehension and processing studies.

4. SUMMING UP: OBSERVATIONS AND FUTURE DIRECTIONS

Though we have sampled only a small fraction of the literature, we stop here and reflect on the themes that have emerged so far. After laying out analytic frameworks for information structure and prosody in Sections 1 and 2, in Section 3 we have reviewed empirical findings that bear on the question of whether and how pitch accenting relates to focus and givenness, yielding four observations:

1. Auditory prominence judgments are highly variable, influenced not only by phonetic cues but also by information from multiple types of context. This challenges the notion that auditory impressions of accent alone are sufficient grounds for a theory of prosody and information structure and highlights the importance of approaches that integrate empirical data related to the production, perception, and comprehension of prosody.
2. Information structure alone is not sufficient to predict the location or type of pitch accent in speech production. Evidence from pitch accents points to a preference for focused words to be accented and given words to be deaccented, though deaccenting is less evident from data on acoustic prominence. And although there is a modest preference for the use of rising nuclear accents with contrastive and narrow focus, the general finding is that speakers produce a variety of pitch accents (or none) in prenuclear and nuclear positions, in all information structure contexts. If pitch accents are pragmatic morphemes, they are optional and not frequently used for this function, especially in laboratory speech elicited from interactive tasks or read text.
3. Prosody has an affordance to cue information structure in speech processing and comprehension. Whether effects are observed or not depends on the task, the salience of prosodic contrasts in the stimuli, and the comparisons presented to listeners. Online tasks are more revealing of pitch accent effects than offline processing, though subject to individual differences, possibly related to skills in pragmatic language use.

4. Prominence mediates the relationship between sound and meaning, which is captured in terms of a mapping that relates acoustic and phonological prominence scales to scales over information structure categories. A word that ranks high on the focus or givenness scale of informativity is likely to be associated with high ranking on acoustic prominence and/or on the prominence scale for pitch accents.

Regarding the information structure distinctions outlined in Section 1, the studies reviewed here are largely consistent with the prosodic encoding of givenness at the phonological level of pitch accent and in gradient acoustic prominence. There is also modest support for prosodic encoding of focus, as modeled in Rooth's (1992, 2016) theory of focus as semantic alternatives, based on production evidence (Section 3.2) and processing evidence (Section 3.3) for a relationship between accent patterns and contrastive focus. Evidence for Rooth's formulation of focus in terms of antecedent, location, and scope parameters is mixed. Comprehension data from Q-A congruence tasks largely support an effect of accenting on each parameter, while production data are inconsistent in showing effects of focus location and scope on the presence and type of accent.

It is proposed here that the variable expression of information structure in phonological or acoustic phonetic properties can be captured by a mapping between prominence scales. This correspondence is not predicted by current formulations of AM theory but can be captured in an approach that integrates the phonological and phonetic properties of prosody. An example of such an approach is the dynamical systems model of English pitch accents proposed by Iskarous et al. (2024). This model generates a continuum of continuous F0 trajectories while also capturing discrete distinctions corresponding to pitch accent types, reflecting the gradient and categorical behavior pitch accents we have observed in this review.

With ongoing rapid advances in experimental methods, it is premature to draw firm conclusions about the link between prosody and information structure. But the observations emerging from this review support an account based on a general notion of prominence and highlight the need for future work that integrates phonetic and behavioral data from diverse tasks to develop explicit models and explanatory theories of pragmatics and prosody.

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